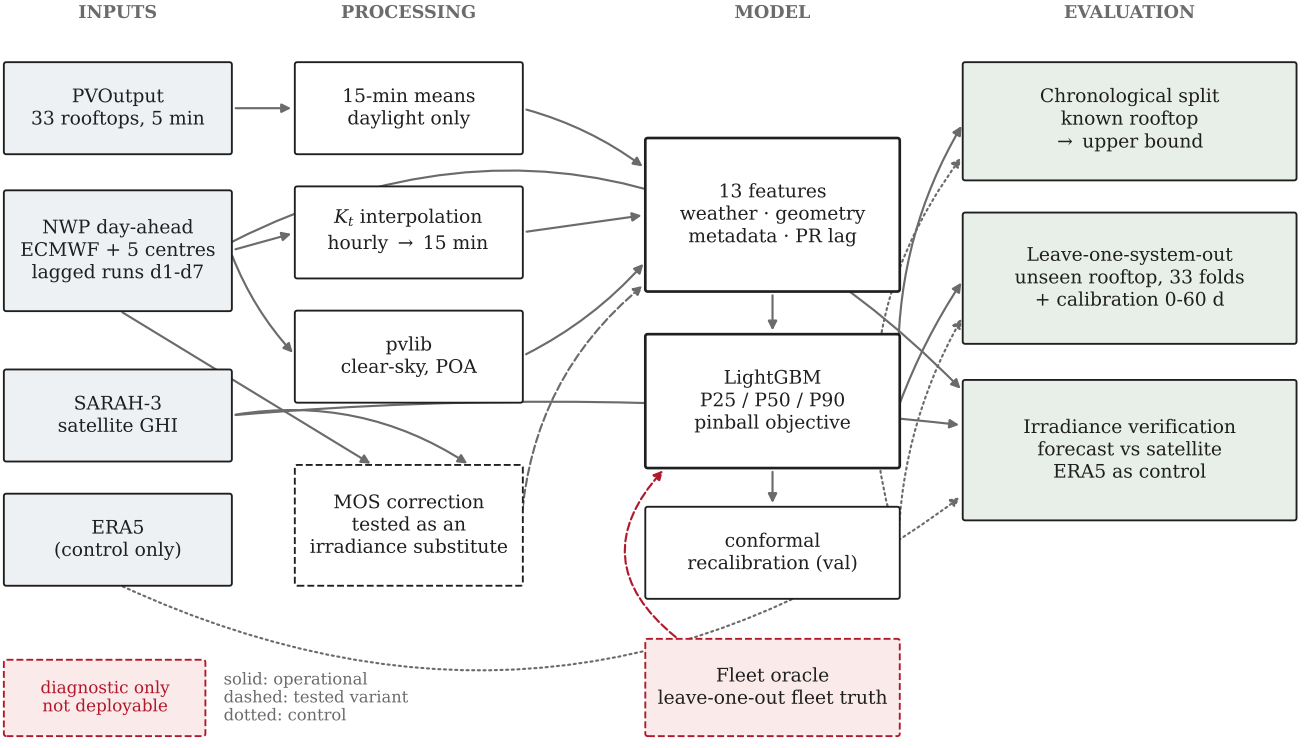
INPUTS
PROCESSING
MODEL
EVALUATION
PVOutput
33 rooftops, 5 min
NWP day-ahead
ECMWF + 5 centres
lagged runs d1-d7
SARAH-3
satellite GHI
ERA5
(control only)
15-min means
daylight only
K_t interpolation
hourly → 15 min
pvlib
clear-sky, POA
MOS correction
tested as an
irradiance substitute
13 features
weather · geometry
metadata · PR lag
LightGBM
P25 / P50 / P90
pinball objective
conformal
recalibration (val)
Fleet oracle
leave-one-out fleet truth
Chronological split
known rooftop
→ upper bound
Leave-one-system-out
unseen rooftop, 33 folds
+ calibration 0-60 d
Irradiance verification
forecast vs satellite
ERA5 as control
diagnostic only
not deployable
solid: operational
dashed: tested variant
dotted: control